Izzac Haug

BACKEND DEVELOPER | SOFTWARE ENGINEER | SYSTEMS PROGRAMMING

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PROFILE & EDUCATION

PROFILE

Entry-level software engineer focused on backend development, automation, and systems programming. Experience contributing to full-stack applications and building command-line software in Python.

EDUCATION

B.S. in Computer Science (Senior)
University of North Carolina at Pembroke
Expected December 2026

TECHNICAL SKILLS

Languages: Python, JavaScript, MQL5, Java, SQL
Technologies: Node.js, React, React Native, Vite, Linux, Git, REST APIs, Ollama, OCR
Focus: Backend Development, Systems Programming, API Design, Automation, Data Processing

PROJECTS

SecretaryAI — AI-Assisted Workflow Platform

Capstone Team Project · In Progress | JavaScript, Node.js, React, Vite, Ollama, OCR, APIs

- Contributed to a full-stack application designed to centralize and automate document workflows for small-business tax agencies.
- Develop and refine backend API routes and application logic supporting document processing and workflow automation.
- Contribute to OCR-based document recognition, document scanning, and database functionality.
- Help integrate and test an Ollama-powered chatbot, troubleshoot bugs, and conduct ongoing backend security reviews.

Custom Shell — Python Command-Line Interpreter

Python · Linux · Git

- Implement a custom shell against a 76-stage workflow, covering built-in commands, external command execution, PATH lookup, quoting, redirection, and pipelines.
- Develop interactive shell features including programmable command completion, file and directory completion, background job management, and persistent command history.
- Implement shell-variable storage and environment-variable expansion; brace expansion and empty-variable expansion remain in development.
- Design the shell as an extensible foundation for future custom built-ins and additional functionality.
- Troubleshoot and fix bugs in the future.

customMQL5 — Algorithmic Trading Engine

MQL5

- Designed and implemented an algorithmic trading engine with modular strategy and risk-management components.
- Built trade execution, indicator analysis, and market-data processing functionality.
- Developing probabilistic learning features to dynamically adjust trading strategies.

theNutritionist — Full-Stack Nutrition Tracking App

React Native · JavaScript · Backend APIs | In Progress

- Develop a cross-platform nutrition tracking app with food logging and goal management.
- Contribute to backend API functionality and application architecture for managing user data.
- Work on authentication and real-time data updates.

GPU-Calc — Crypto Mining Performance Calculator

Backend Services · Deployment Infrastructure | Contributor

- Implemented server-side logic for processing hardware performance data in a calculator application.
- Configured HTTPS, domain settings, and deployment infrastructure.
- Contributed to backend services supporting the application's performance calculations.